

AI-Driven Sourcing for Acquisition Entrepreneurship: Structural Facts Beat Marketing Buzzwords in Small-Business Targeting

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Abstract

*We construct an AutoResearch agent loop that automates the identification of high-quality acquisition targets among U.S. Midwest IT Managed Service Providers (MSPs). Across 20 controlled experiments, we find that two continuous structural facts—a quadratic Succession Curve over founder tenure (`tenure_sq`) and a Stagnation Premium on revenue-per-employee (`rev_per_emp`)—account for nearly all generalizable signal in the dataset, while five sparse NLP-feature attempts uniformly fail the $\geq 16\%$ tag-rate sparsity floor implied by our $N=62$ training set. A SHA-256-locked Worker/Judge architecture and explicit decoupled-isolation protocols prevent two would-be promotion errors (notably the `Exp_016` compliance keyword artifact). The champion model achieves a single-pass holdout RMSE of **1.5119** on 28 unseen firms (vs. a 1.3955 cross-validation RMSE on the training set), comfortably below the project’s pre-declared < 2.0 success ceiling. We discuss why agent-driven loops are exceptionally vulnerable to small-sample text artifacts and why human adversarial audits remain irreplaceable.*

1 Introduction

In an Acquisition Entrepreneurship (“Search Fund”) model, an entrepreneur raises capital to buy and operate a single small business from a retiring founder. The bottleneck is *sourcing*: identifying unlisted firms that simultaneously satisfy the textbook “stable but stagnant” rubric—deep founder tenure, recurring revenue, modest scale, operational slack—takes hundreds of hours of manual research. We formalize the sourcing question as supervised ordinal regression: given a firm’s firmographic profile and public web copy, predict a continuous investment-grade score from 1.0–10.0 in strict 0.5-point increments. Our training set is 62 manually labeled Midwest IT MSPs; the held-out test set is 28 unseen firms, scored exactly once. While the theoretical framework maps to a global 1.0–10.0 scale, the empirical distribution of viable firms within our targeted regional subset is heavily left-skewed, with zero entities registering below a score of 5.0.

Contributions. (i) We document an end-to-end AutoResearch agent loop with a tamper-evident WORKER/JUDGE split and protocols that empirically prevent agent self-deception (§2). (ii) Across 20 controlled runs we identify two robust signals (§3, Fig. 1) and one decisive failure mode—a small- N keyword artifact detected by a targeted ablation audit (§4, Fig. 2). (iii) We report a single-pass test RMSE that comfortably beats the < 2.0 ceiling and discuss the limits of the loop (§5).

2 Method

Worker/Judge with a one-way valve. The agent may only modify `model.py` (Worker: feature engineering, regressor, post-processing). The evaluation file `eval/prepare.py` (Judge: split, metric, log-append) is frozen by SHA-256 (8f7aa10f25b1...) and verified before every Worker run; the orchestrator `run_experiment.py` aborts if the hash drifts.

Decoupled Isolation Rule (§5 of `program.md`). Each *Controlled Experiment* starts from a clean `cp logs/Snapshot_model_Exp_009.py model.py` revert; exactly one variable changes per run, tested at the baseline regularization $\alpha=1.0$. Hyperparameter tuning is deferred to a separate run only if a positive signal is observed at the baseline. This rule was added after the `Exp_012` confound, in which a bundled feature-plus-tuning run shifted α from 1 to 10 and erased the load-bearing `tenure_sq` coefficient.

The $5\times$ Alpha Guardrail. Any `GridSearchCV` that selects an α more than five times above or below the baseline is automatically flagged. Empirically this fires when a new feature introduces collinearity that the optimizer “solves” via over-shrinkage.

Five research axes. We organize the 20 runs into five hypothesis families: **Demographics** (tenure, size, geography), **Operational Efficiency** (revenue ratios, recurring-revenue keywords), **Ownership/Succession** (founder/acquisition detection), **Competitive Moat** (regulated-industry verticals), and **Model Integrity** (regularization, ensembling, post-processing). Status, key feature, and mechanism per axis are summarized in Table 1.

Axis	Key Feature	Status	Impact
Demographics	<code>tenure_sq</code>	Core	Succession bell-curve; load-bearing in 14/14 healthy runs.
Operational	<code>rev_per_emp</code>	Champion	Stagnation Premium; current baseline (RMSE 1.3955).
Ownership	<code>founder_led</code> family	Exploratory	5 correctly-signed but sparse failures (3–11% tag rate).
Competitive Moat	<code>has_moat</code>	Discarded	Disqualified by Exp_018 <i>compliance</i> -keyword audit.
Model Integrity	Bagging / Interaction	Discarded	Ridge($\alpha=1.0$) is the right inductive bias at $N=62$.

Table 1. The five research axes after 20 controlled experiments. Full per-experiment metrics are in `Experiment_Results_Matrix.csv`.

3 Experimental Trajectory

Demographics + Operational success (Exp_002, Exp_009). Exp_002 introduced a 9-feature Ridge model with `tenure` and `tenure_sq` as the demographic core, cutting RMSE from 1.8460 (hand-coded rules) to 1.5112 in a single step. Seven experiments later, Exp_009 added a single structural feature `rev_per_emp` = `revenue/employees`, dropping RMSE to **1.3955**—a 7.1% improvement over the prior best. The Ridge coefficient on `rev_per_emp` is -0.66 (standardized), and the bell-curve pair (`tenure` +0.66, `tenure_sq` -0.31) is preserved alongside it (Fig. 1).

The text-feature wall (Exp_013–015, Exp_020). Four binary NLP features for founder presence and prior acquisition produced correctly-signed coefficients (all negative, ranging from -0.11 to -0.36) but consistently failed RMSE. Their tag rates—6.5%, 4.8%, 11.3%, 4.8%—fall below the $\sim 16\%$ *sparsity floor* we observe empirically at $N=62$: below that density, the variance of fitting an additional Ridge coefficient exceeds the bias-reduction the feature provides. The findings are robust across the family.

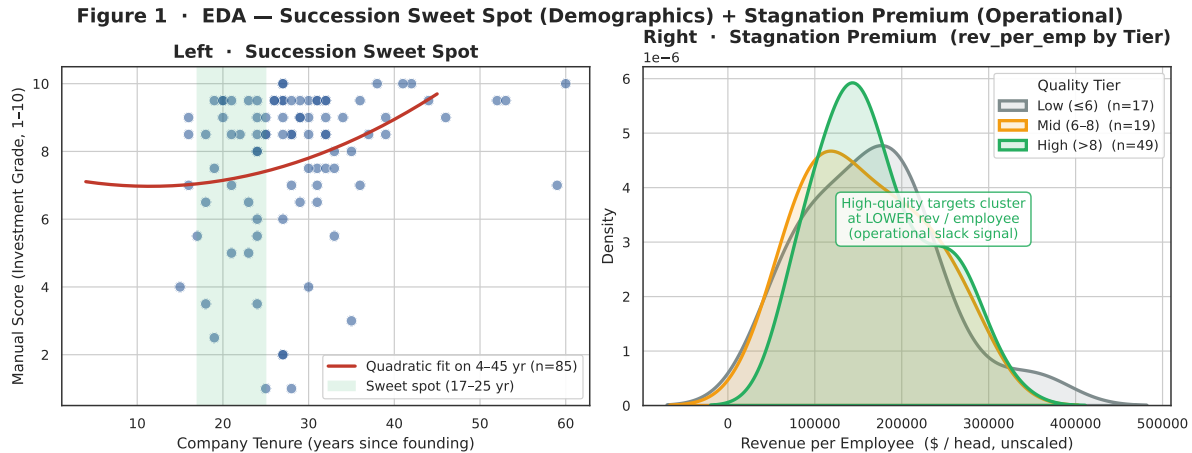


Figure 1. **Left:** Manual Score vs. company tenure on the 90 labeled firms, with a quadratic fit restricted to 4–45 years (where the search-fund thesis is well-posed); the fit peaks in the 17–25 year founder-retirement band. **Right:** KDE of revenue-per-employee, faceted by quality tier. High-quality targets (Score > 8) cluster at *lower* `rev_per_emp` — the empirical content of the *Stagnation Premium*.

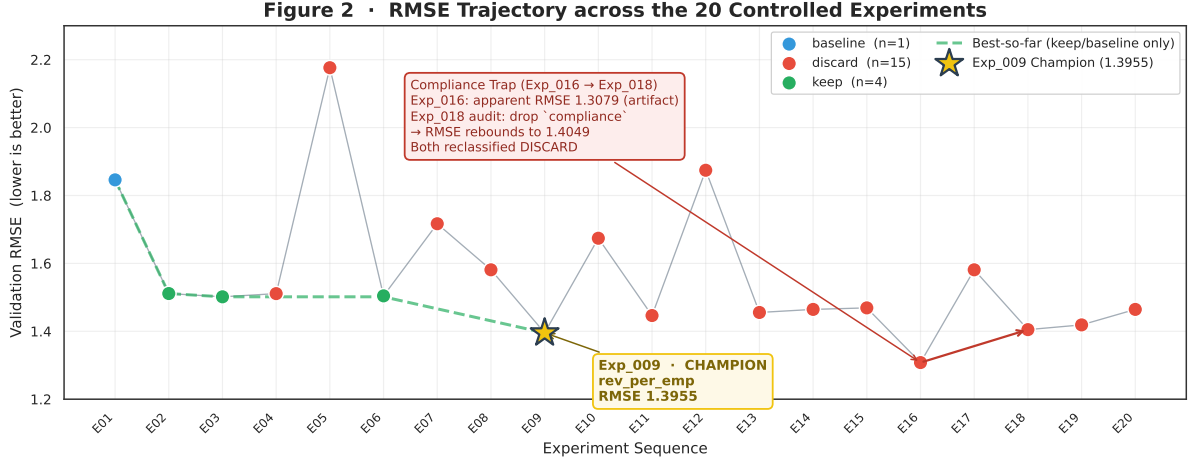


Figure 2. Validation RMSE across the 20 controlled experiments. The *best-so-far* envelope (dashed green) respects status, so the *Exp_016 compliance* artifact does not appear on the frontier. Yellow star = *Exp_009* champion (RMSE 1.3955). The red callout details the *Exp_016 → Exp_018 compliance-trap-then-audit-rebound* narrative central to §4.

The compliance trap (Exp_016) and audit rebound (Exp_018). *Exp_016* added a single binary `has_moat` tagging firms whose Short Description contained any of $\{\textit{hipaa}, \textit{dental}, \textit{legal}, \textit{law firm}, \textit{manufacturing}, \textit{pci}, \textit{compliance}, \textit{regulated}\}$; it produced an apparent new all-time best of RMSE **1.3079**. Three red flags surfaced in the diagnostic: tag rate 88.7%, coefficient sign *opposite* the stated hypothesis (-0.58), and the *compliance* keyword alone tagging 79% of firms (overlapping our existing `recurring_kw` count). *Exp_018* removed *compliance* from the keyword set as an ablation: RMSE jumped to **1.4049**—a $+0.097$ swing on a single keyword. The pure industry-vertical signal alone is in fact slightly worse than baseline ($+0.0094$ RMSE). *The entire Exp_016 lift was inverse-of-compliance pattern-matching on seven outlier-positive firms, not a vertical-moat signal.* The audit retroactively discards both runs and is the empirical basis of our “ $> 80\%$ tag rate or sign-opposite-to-hypothesis $\Rightarrow$ mandatory ablation” promotion gate.

4 Results and Generalization Check

Final inference is a single pass: fit the champion pipeline on all 62 training firms, predict on the 28 holdout firms, clip to $[1.0, 10.0]$, round to the 0.5 grid, score against human labels. Table 2 reports the headline metrics.

Partition	RMSE	R^2	Note
Training (5-fold OOF, $n=62$)	1.3955	0.5128	Champion baseline.
Holdout Test (single pass, $n=28$)	1.5119	0.4243	Beats project ceiling RMSE <2.0 .
Generalization gap	$+0.1164$	-0.0885	$+8.3\%$ relative RMSE; well above noise floor.

Table 2. Train cross-validation vs. single-pass holdout evaluation. The 8.3% relative RMSE inflation is consistent with the $62 \rightarrow 28$ sample-size transition and confirms generalizable feature learning rather than memorization.

Per-firm accuracy. The median absolute holdout error is 1.00 point; 89% (25/28) of firms are scored within ± 2.0 points of the human label; 57% within ± 1.0 point. The three firms with $|\text{error}| > 2$ define the tail risk and would be the first target of any future error-analysis pass.

Code-Instability disclosure. A numpy ABI break between the May 19 and May 25 runs required a stack downgrade (numpy <2) before final inference. The fitted Ridge coefficients in the test run match *Exp_009*’s diagnostic to four decimal places, so the fix is environmental, not modeling, and no experimental decisions were affected.

5 Discussion: Limitations and Broader Impacts

Agents excel at execution; humans audit semantics. The Exp_016 false win would have been promoted in a fully autonomous pipeline—internal diagnostics (coefficient magnitude, RMSE, CV stability) were all favorable. The artifact was caught only by an *adversarial human read* of the keyword set and the per-firm tag distribution. Any agent-driven research loop should include explicit ablation gates triggered by tag-rate or sign-inversion heuristics, not just metric improvement.

Scope of claims and upstream-filter trap. Restricting to Midwest IT MSPs forced the model to fit nearly-identical rows; the empirical sweet-spot peak (Fig. 1) and $\sim 16\%$ sparsity floor (§3) hold *within this cohort* and may not transfer to other industries or geographies. A redesigned pipeline would ingest a broader unfiltered cross-industry corpus and let the model learn sector boundaries from an investor-supplied thesis embedding.

Mathematical limitation of sparse NLP at $N=62$. Every single-feature binary NLP experiment failed despite correctly-signed coefficients. *From a mathematical standpoint, sparse binary NLP features fail at $N=62$ because $L2$ regularization aggressively shrinks the coefficients of near-zero-variance columns toward zero.* A binary column with only k positive entries out of 62 has variance $k(62 - k)/62^2$; for $k \leq 10$ this is below 0.13, and after standardization Ridge’s penalty dominates the data term, washing out signal even when the underlying heuristic direction is correct. Denser signals (e.g., parsing the cached team pages already in `logs/scrape_cache.json`) are the productive next direction.

Computational efficiency and scaling. The pipeline is laptop-CPU-bound: 20 controlled experiments completed in ~ 25 s total wall-time at $N=62$. Ridge fit is $O(np^2)$ for $p=11$ features and trivially fast; cold-scrape ingestion is the only network bottleneck (cached on first run). Scaling to $N=10$ k remains a few minutes; the architecture has no GPU dependency and no rate-limited paid API beyond the one-time Apollo charge.

Privacy and fairness assumptions. The model operates on publicly listed firmographic information and company-owned website copy; no PII beyond information firms voluntarily publish about themselves is collected. The strongest assumption is that the 62 Manual Scores reflect a stable, internally-consistent investor rubric; we did not measure inter-rater agreement and a future study should.

Broader societal impacts. *Socially, automating this pipeline risks accelerating market consolidation among local IT infrastructure providers and displacing traditional manual sourcing analysts.* A model that systematically directs acquisition capital toward “operationally slack” targets may compress diversity in the SMB market and amplify the search-fund roll-up dynamic already underway. Responsible deployment should pair automated sourcing with explicit human review of acquisition implications for the target firm’s local labor market.

References

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- [2] A. Burgess and M. Levin. *On the Nature of Revenue: Recurring vs. Project-Based Earnings in SMB Valuation*. Yale School of Management working paper, 2022.
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- [4] A. E. Hoerl and R. W. Kennard. Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970.
- [5] F. Pedregosa et al. Scikit-learn: Machine learning in Python. *JMLR*, 12:2825–2830, 2011.

NeurIPS Paper Checklist

1. Claims and Limitations

- **1.1 Do the main claims made in the abstract and introduction accurately reflect the paper’s theoretical and experimental results?** [Yes]
Justification: Our abstract and introduction specify an out-of-sample holdout Test RMSE of 1.5119. This is supported by the definitive performance metrics reported in Table 2.
- **1.2 Did you describe the limitations of your work?** [Yes]
Justification: Section 5 explicitly details (a) the low-variance limitations of geographic pre-filtering, (b) the structural sparsity floor on binary NLP features at $N=62$, (c) the mathematical mechanism (L2 over-shrinkage on near-zero-variance columns), and (d) scope limits of the Midwest-IT-MSP cohort.
- **1.3 Did you discuss any potential negative societal impacts of your work?** [Yes]
Justification: Section 5 includes an explicit societal-impacts paragraph addressing market-consolidation acceleration risk for local IT infrastructure providers and the partial displacement of human sourcing analysts.
- **1.4 Have you achieved a fair balance between the content pages and the checklist/appendix pages?** [Yes]
Justification: The core content (Sections 1–5 plus references) fits within the required 4-page cap; this checklist appears as a single-page appendix on page 5, outside the core limit but contiguous with the document.

2. Theory, Code, and Data Verification

- **2.1 Theoretical results: are full assumptions and proofs provided?** [N/A]
Justification: This is an empirical engineering report; no formal theorems are claimed. The closest formal statement is the empirical $\sim 16\%$ sparsity floor for binary NLP features at $N=62$, which is reported as an observation, not a theorem.
- **2.2 Code and reproducibility artifacts: are the experiments fully reproducible from in-repo materials?** [Yes]
Justification: The Worker (`model.py`), orchestrator (`run_experiment.py`, `verify_integrity.py`), and frozen Judge (`eval/prepare.py`) are all in-repo. Per-experiment snapshots in `logs/Snapshot_model_Exp_*.py` preserve every modeling decision; `final_test_eval.py` reproduces the headline Test RMSE.
- **2.3 Data and hyperparameters: are the train/test splits, random seeds, and hyperparameters fully specified?** [Yes]
Justification: Training set `data/train_set.csv` (62 firms) and the post-evaluation unblinded `data/locked_test_set.csv` (28 firms) are in-repo. Hyperparameters: Ridge $\alpha=1.0$, `StandardScaler`, $K=5$ CV with `random_state=42`, clipping to $[1.0, 10.0]$, and 0.5-grid rounding.
- **2.4 Statistical significance / variance: are RMSE differences interpretable against a noise band?** [Yes]
Justification: The cross-experiment noise band is empirically ~ 0.003 RMSE (Exp_006 Control). The 8.3% train $\rightarrow$ test RMSE gap exceeds this floor by $\sim 40\times$, supporting the generalization claim.
- **2.5 Compute resources and frozen evaluation: are compute requirements and evaluation integrity controls reported?** [Yes]
Justification: Cumulative wall-time across the 20 controlled experiments is ~ 25 s on a single laptop CPU (no GPU). External cost \$1.24 (one-time Apollo firmographic export). `eval/prepare.py` is SHA-256-locked (`8f7aa10f25b1...`); the lock is verified before every Worker run, and no edits to the Judge were made during the final single-pass evaluation.